

Smart Trolley Robot 'TROLL.E 1.0'

NAVPREET SINGH TUNG

Robots now play a vital role across modern society, often described as human-like due to their growing presence in social and commercial environments. With advances in artificial intelligence, they are increasingly seen as capable of exhibiting almost human-level intelligence.

Robots' applications span across diverse domains—industry, homes, hospitality, events, agriculture, and even space exploration. Within this broad landscape, the Smart Trolley Robot, TROLL.E 1.0, is being introduced here. Lightweight, structurally

simple, and engineered for precise control, the robot offers both efficiency and ease of use.



Fig. 1: Author's prototype

In most malls, shoppers continue to rely on mechanical trolleys that require manual effort. Smart electric alternatives remain uncommon in retail and warehouse settings. Although some digital trolley models exist, they tend to be expensive and overly complex. Fig. 1 shows author's prototype for a simple yet smart trolley robot.

To assemble this robot, the required components are listed in the Bill of Materials table.

Circuit and working

Fig. 2 shows the circuit diagram of the smart trolley robot. Bluetooth

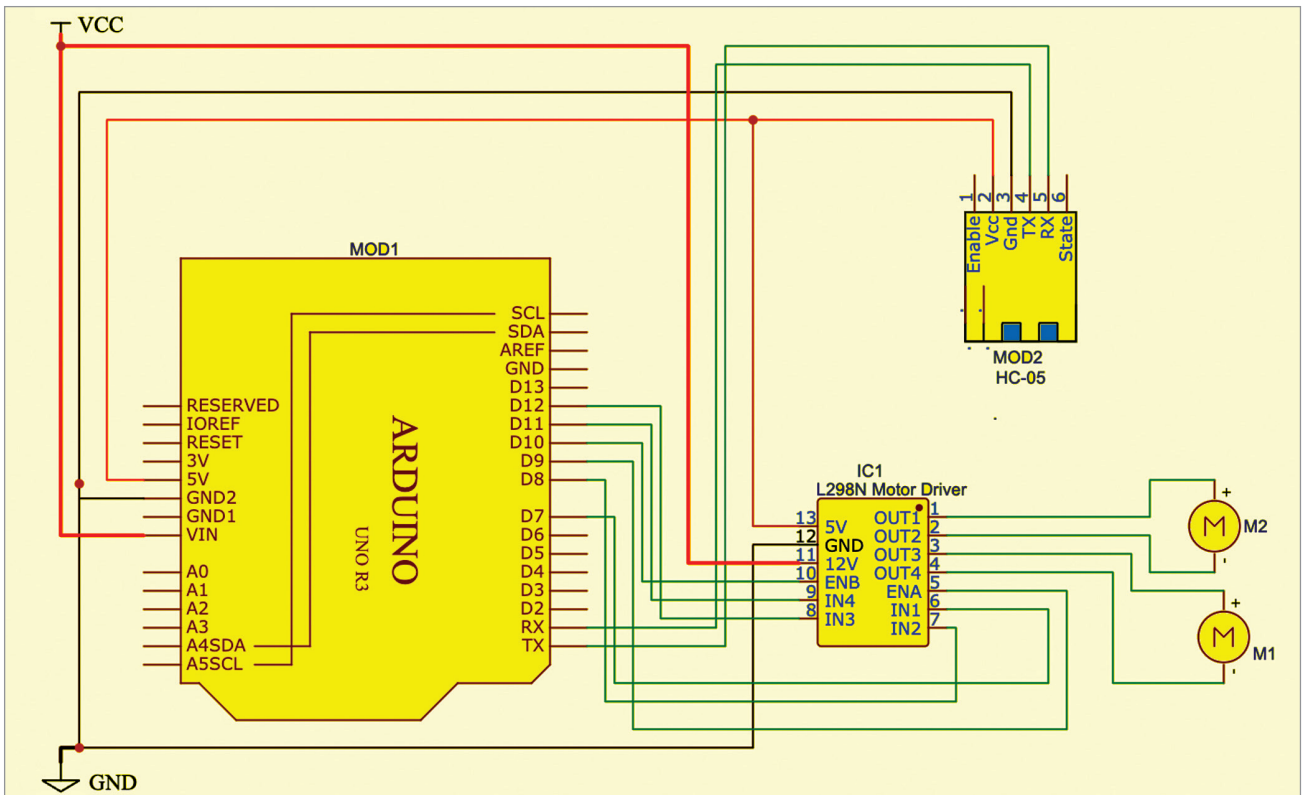


Fig. 2: Circuit diagram